American Computer Science League

2024-2025 ● Contest 1: Short Problems ● Elementary Division

1. Computer Number Systems

Convert 2024_8 to base 10.

A. 8
B. 532
C. 1022
D. 1044
E. 2178

2. Computer Number Systems

Convert $A27_{16}$ to base 8.

A. 537
B. 547
C. 5047
D. 5057
E. 6037

3. Computer Number Systems

What is the number 198_{10} in base 16?

A. 126
B. 6C
C. B6
D. C5
E. C6

4. Computer Number Systems

How many 1's are there in all of the binary numbers from 20_{10} to 24_{10} inclusive?

A. 13
B. 14
C. 15
D. 16
E. 17

5. Computer Number Systems

Given the two colors $B0E7C3_{16}$ and $98FDAE_{16}$, what is the largest positive difference between the shades of red, green, or blue in hexadecimal?

A. 16
B. 17
C. 18
D. 19
E. 20

6. Computer Number Systems The colors for the Olympic Rings are 0081C8 (water blue), 008651 (shamrock green), 000000 (black), FCB131 (macaroni and cheese), and EE334E (dark carmine pink), how many of the 5 colors do not have a red component?	A. 0 B. 1 C. 2 D. 3 E. 4

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1. 计算机数字系统

将 2024_8 转换**为** 10 进**制。**

A. 8
B. 532
C. 1022
D. 1044
E. 2178

2. 计算机数字系统

将 $A27_{16}$ 转换为 8 进**制。**

A. 537
B. 547
C. 5047
D. 5057
E. 6037

3. 计算机数字系统

198_{10} 用 16 进**制表示是什么**?

A. 126
B. 6C
C. B6
D. C5
E. C6

4. 计算机数字系统

当 20_{10} 到 24_{10} 之间**（包括** 20_{10} **和** 24_{10}**）**的所有数用**二进制**表示时，总共**有多少个** 1**？**

A. 13
B. 14
C. 15
D. 16
E. 17

5. 计算机数字系统

给定**两种颜色** $B0E7C3_{16}$ **和** $98FDAE_{16}$，分别计算这两种颜色的**红、绿、蓝**色阶之间的差值，并用**十六进制表示**。请问其中**最大正**数差**是多少？**

A. 16
B. 17
C. 18
D. 19
E. 20

6. 计算机数字系统 奥运**五环的颜色包含** 0081C8（水蓝）、008651（三叶草**绿**）、000000（黑**色**）、FCB131（类似芝士通心粉的**黄色**）、EE334E（深**粉红**），请问**这** 5 **种颜色中有几种颜色不含红色**阶？	A. 0 B. 1 C. 2 D. 3 E. 4